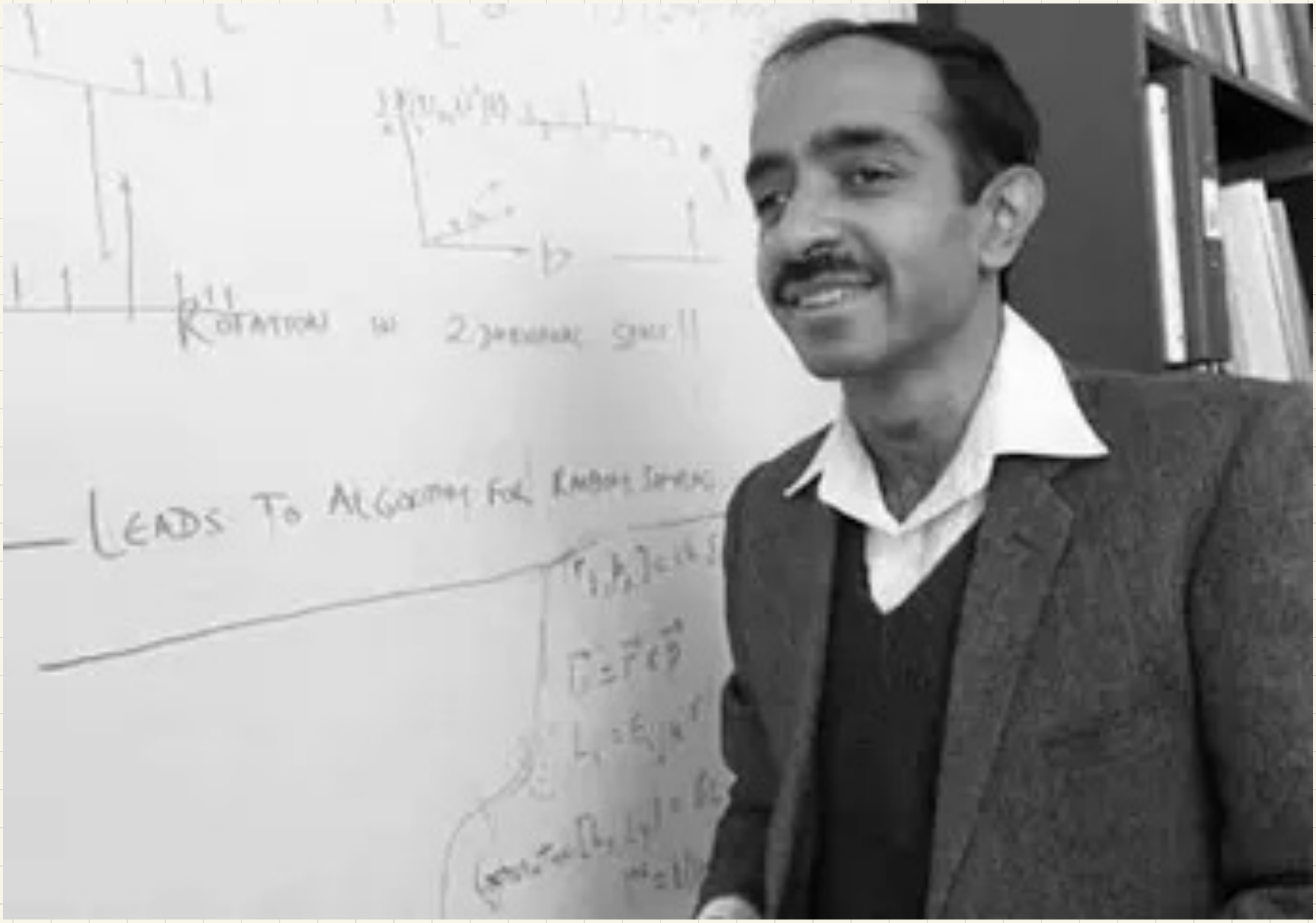
Lecture 04/04/2025



Searching in unstructured database

$X = \{|0\rangle, |1\rangle \dots |N-1\rangle\}$ $\quad |i\rangle$ is in X

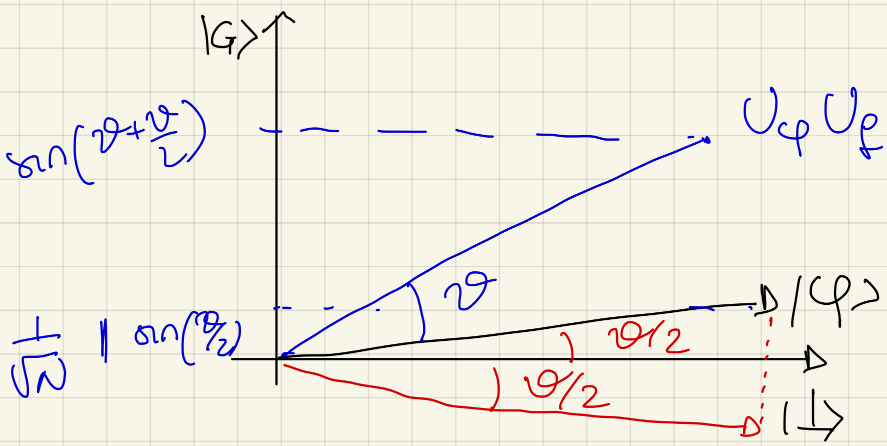
$$f(x) = \begin{cases} 0 & \text{if } x \text{ is not a solution } \quad x \notin G \\ 1 & \text{if } x \text{ is a solution } \quad x \in G \end{cases}$$

1. $U_f|i\rangle = (-1)^{f(i)}|i\rangle$ Oracle

2. $U_\varphi = 2|\varphi\rangle\langle\varphi| - I$

$$|G\rangle = \frac{1}{\sqrt{t}} \sum_{i:\, f(i)=1} |i\rangle \qquad |\perp\rangle = \frac{1}{\sqrt{N-t}} \sum_{i:\, f(i)=0} |i\rangle$$

$$|\varphi\rangle = \frac{1}{\sqrt{N}} \sum_{i=0}^{N-1} |i\rangle = \sqrt{\frac{t}{N}}\, |G\rangle + \sqrt{\frac{N-t}{N}}\, |\perp\rangle$$



$$U_f|\varphi\rangle = -\sqrt{\frac{t}{N}}\,|G\rangle + \sqrt{\frac{N-t}{N}}\,|\perp\rangle$$

k iterations $\rightarrow$ $\sin\left(k\vartheta + \frac{\vartheta}{2}\right) \simeq 1$

$$k\vartheta + \frac{\vartheta}{2} \approx \frac{\pi}{2}$$

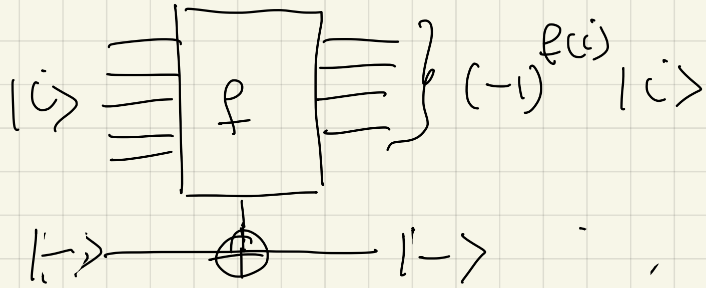
$$k = \left\lfloor \frac{\pi}{4}\sqrt{N} \right\rfloor$$

$\Pr(\text{measuring } |G\rangle) > 1 - \frac{1}{N}$ if $t = 1$, $k = \left\lfloor \frac{\pi}{4}\sqrt{N} \right\rfloor$

How to construct the oracle?

If you have a classical circuit for

$f: \{0,1\}^n \rightarrow \{0,1\}$ $\qquad f(i) = \begin{cases} 0 & \text{if } i \in G \\ 1 & \text{if } i \notin G \end{cases}$



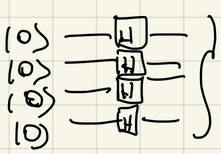
ex: given a large m if $m = p \cdot q$

Check if $p \cdot q = m$ easy, finding p and q is difficult.

The Diffuser.

$$U_\varphi = 2|\varphi\rangle\langle\varphi| - I$$
$$= 2\, H^{\otimes n} |0\rangle^{\otimes n} \langle 0|^{\otimes n} H^{\otimes n} - H^{\otimes n} H^{\otimes n}$$
$$= H^{\otimes n} \big(2|0\rangle\langle 0| - I \big) H^{\otimes n}$$

$\uparrow$ n qubit



$$|\varphi\rangle = H^{\otimes n} \left(\begin{bmatrix} 2 & & 0 \\ & 0 & \\ 0 & & \ddots \\ & & & 0 \end{bmatrix} - I \right) H^{\otimes n}$$

$$= H^{\otimes n} \underbrace{\begin{pmatrix} 1 & & & \\ & -1 & & \\ & & \ddots & \\ & & & -1 \end{pmatrix}}_{= U_0} H^{\otimes n}$$

$$U_0 = - X^{\otimes n} (MTG)\, X^{\otimes n}$$

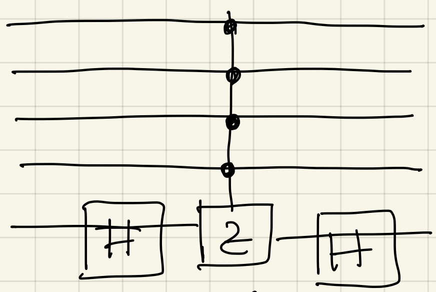
$\uparrow$ multi-Toffoli Gate

$$-X^{\otimes n}\begin{bmatrix}1&&&\\&-1&&\\&&\ddots&\\&&&-1\end{bmatrix}X^{\otimes n} = MTG$$

$$\Rightarrow -MTG = \begin{bmatrix}1&&&&\\&\ddots&&&\\&&1&&\\&&&\underbrace{\begin{matrix}1&\\&-1\end{matrix}}_{C-Z}\end{bmatrix}$$

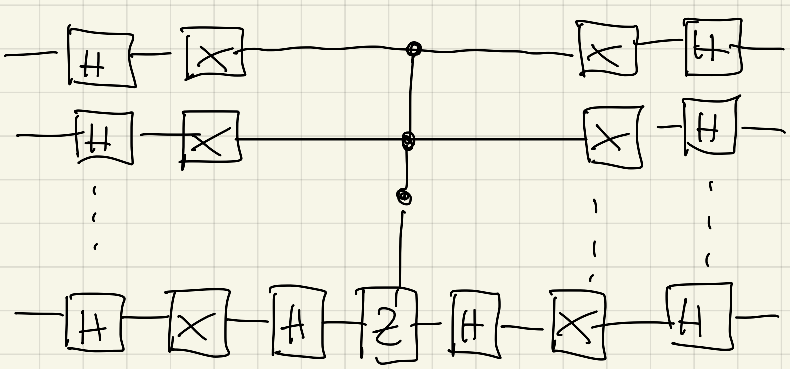
$$Z=\begin{bmatrix}1&\\&-1\end{bmatrix}$$

$$X=\begin{bmatrix}0&1\\1&0\end{bmatrix} \qquad X\otimes X = X^{\otimes 2} = \begin{bmatrix}0&0&0&1\\0&0&1&0\\0&1&0&0\\1&0&0&0\end{bmatrix}$$

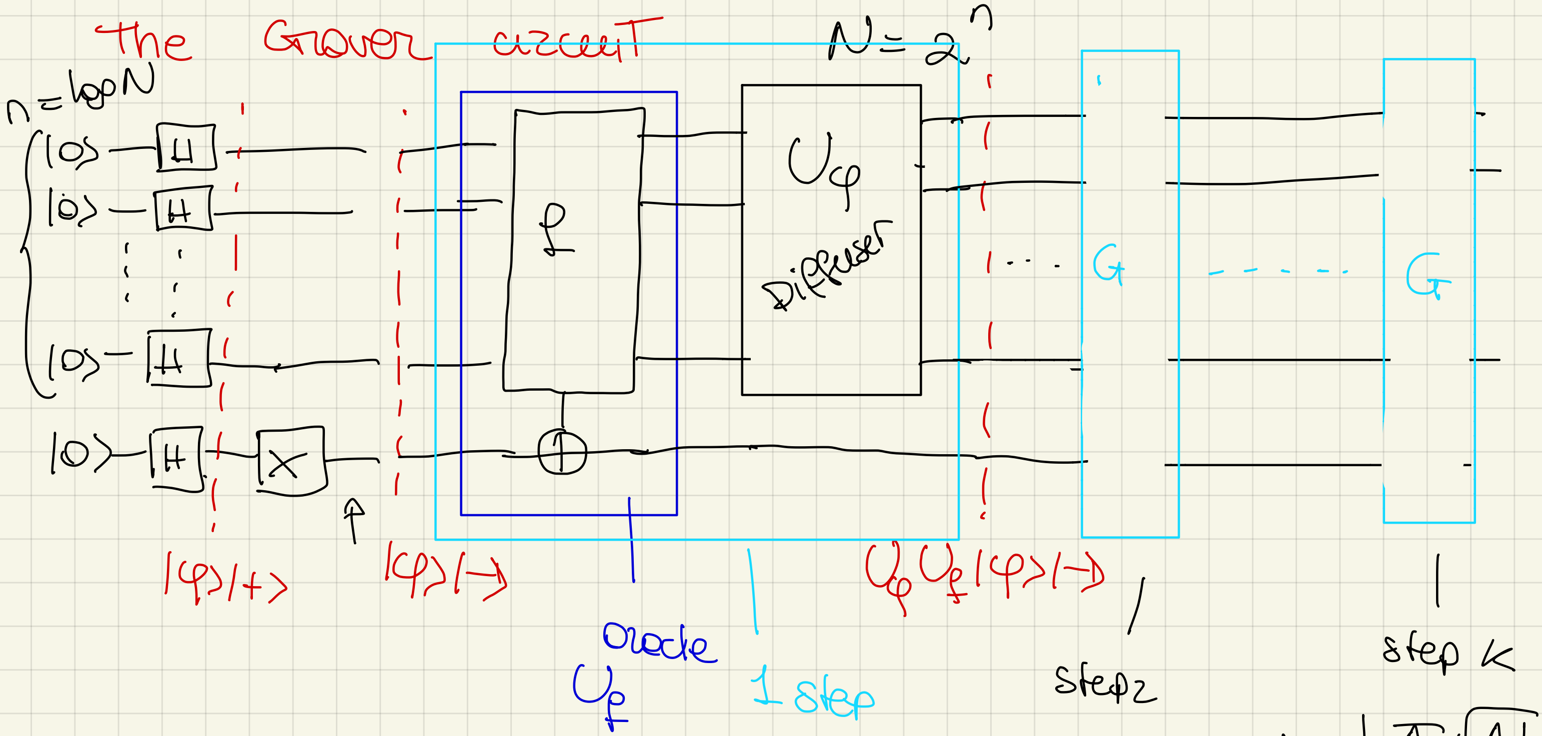


$-MTG$

the circuit for the diffuser is



The Grover circuit



$$k = \left\lfloor \frac{\pi}{4}\sqrt{\frac{N}{t}} \right\rfloor$$

Applications of Grover's search idea

$$2^n \to 2^{n/2}$$

Grover algorithm is however optimal for searching

$$\Omega(\sqrt{N})$$

MAXIMIZATION PROBLEMS

$$f : \overbrace{\{0, 1, \dots, N-1\}}^{X} \to \mathbb{R}_+$$

find i such that $f(i) = \max_{x \in X} f(x)$

Classically you need $O(N)$ by brute force

(Durr – Hoyer) using Grover you can reduce the cost $O(\log N \sqrt{N})$ $\rightsquigarrow$ improved to $O(\sqrt{N})$

1. $\mu = 0$ & keep the maximum

2. repeat

   Use Grover for the following search problem

   $x \in \{0,1\}^n$ $\qquad x_i = \begin{cases} 1 & \text{if } f(i) > \mu \\ 0 & \text{otherwise} \end{cases}$ $\quad \mu = f(i)$

computing x_i with a quantum circuit is easy

- compute $f(i)$
- compare with μ
- output 1 if $f(i) > \mu$, otherwise 0
- uncompute $f(i)$

CLAIM: After $O(\log N)$ repetitions (binary search) we have found the maximum

$$O(\log N \sqrt{N})$$

Amplitude Amplification

You have an algorithm A such that

$$A|0\rangle^{\otimes n} = \sqrt{p}\,|\varphi_1\rangle + \sqrt{1-p}\,|\varphi_0\rangle$$

We assume $\quad \langle\varphi_0|\varphi_1\rangle = 0$

$$\langle\varphi_0|\varphi_0\rangle = \langle\varphi_1|\varphi_1\rangle = 1$$

In the Grover's algorithm

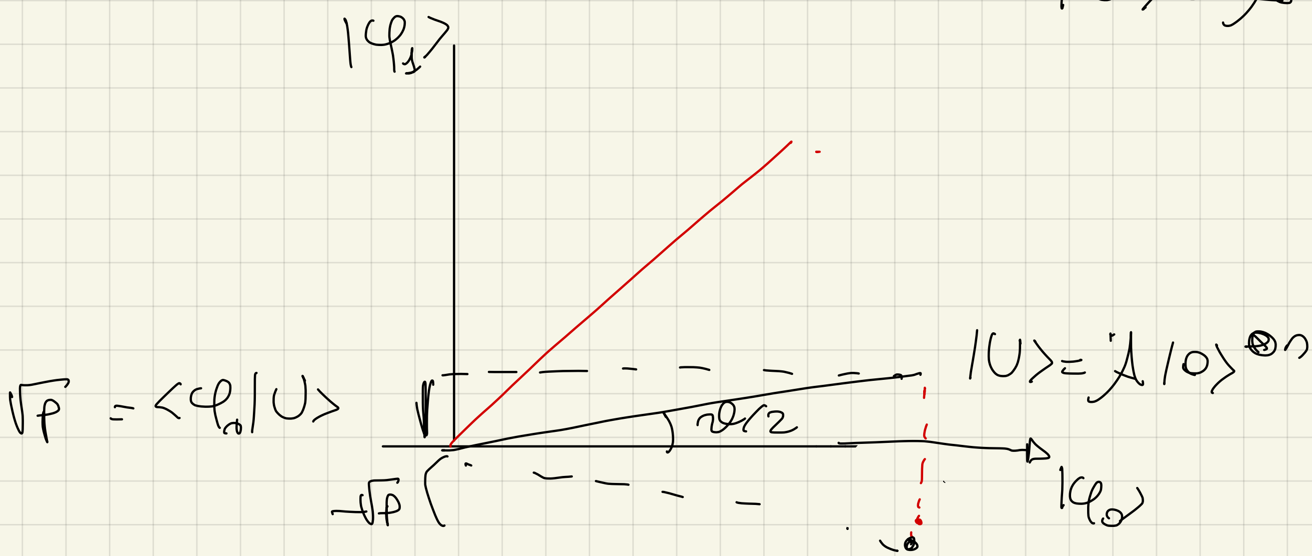
$$\mathcal{A} = H^{\otimes n}$$

$$|\varphi_1\rangle = |G\rangle$$

$$|\varphi_0\rangle = |\perp\rangle$$

- We do Grover $\rightarrow$ 1. Rotate respect to $|\varphi_0\rangle$

  2. Rotate respect to $|U\rangle = \mathcal{A}|0\rangle^{\otimes n}$



$$\langle\varphi_1|U\rangle = \langle\varphi_1|\left(\sqrt{p}\,|\varphi_1\rangle + \sqrt{1-p}\,|\varphi_0\rangle\right) =$$

$$= \sqrt{p}\,\underbrace{\langle\varphi_1|\varphi_1\rangle}_{1} + \sqrt{1-p}\,\cancel{\langle\varphi_1|\varphi_0\rangle} = \sqrt{p}$$

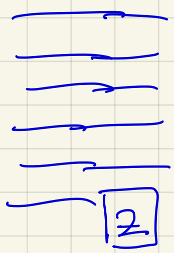
1. $|U\rangle \rightarrow -\sqrt{p}\,|\varphi_0\rangle + \sqrt{1-p}\,|\varphi_1\rangle$

For step 1 we need an oracle R_G for changing the sign to entries in the direction of $|\varphi_0\rangle$

sometime

$$|\varphi_1\rangle = |\ldots\ldots 1\rangle \qquad |\varphi_0\rangle = |\ldots\ldots 0\rangle$$

$$|U\rangle = A|0\rangle^{\otimes n}$$



$$|U\rangle = \sqrt{p}\,|\varphi_1\rangle + \sqrt{1-p}\,|\varphi_0\rangle$$

$$Z = \begin{bmatrix}1 & \\ & -1\end{bmatrix} \quad Z|1\rangle = Z\begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}0\\-1\end{pmatrix} = -\begin{pmatrix}0\\1\end{pmatrix} = -|1\rangle$$

$$Z|0\rangle = Z\begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}1\\0\end{pmatrix} = |0\rangle$$

In this case the oracle is very easy

$$R_G = I^{\otimes n-1} \otimes Z$$

the Diffuser.

$$R_U = 2|U\rangle\langle U| - I$$

$$|U\rangle = A|0\rangle^{\otimes n}$$

$$R_U = A\left(2\overset{U_0}{|0\rangle\langle 0|} - I\right)A^{\dagger}$$

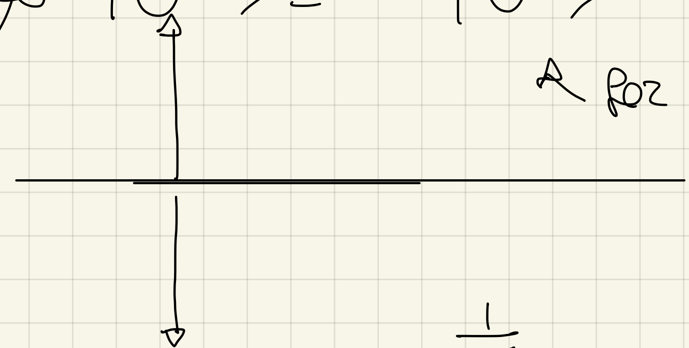
$$R_u|U\rangle = A\, U_0\, A^\dagger |U\rangle = A\, U_0\, \underbrace{A^\dagger A}\, |0\rangle^{\otimes n} = A\, \underbrace{U_0 |0\rangle^{\otimes n}}$$

$$\begin{bmatrix} 1 & & & \\ & -1 & & \\ & & \ddots & \\ & & & -1 \end{bmatrix} |0\rangle^{\otimes n}$$

$$= A\, |0\rangle^{\otimes n} = |U\rangle$$

$$R_u|U^\perp\rangle = A\, U_0\, A^\dagger\, |U^\perp\rangle = -\,|U^\perp\rangle$$

for exercise



AA algorithm

$\frac{1}{\sqrt{N}}$

$\sqrt{p}$

1. set $|U\rangle = A\,|0\rangle^{\otimes n}$
2. Repeat $O\left(\frac{1}{\sqrt{p}}\right)$ times
   - Reflect through bad slot $|e_0\rangle$ — R_a
   - Reflect about $|U\rangle$ — R_U

If we do not know $\sqrt{p}$
we try with exponentially decreasing value of p

$p = \frac{1}{2}$ $\quad p = \frac{1}{4}$ $\quad p = \frac{1}{8}$ ...